

RESUME

RESUME — HelixCode Session Handoff

Revision: 4 **Created:** 2026-07-08 **Last modified:** 2026-07-27 **Status:** active

Maintainer: CLI-agent main work stream **Authority:** §11.4.131 Session-resumption file — point a fresh agent here. Composes §11.4.127 / §12.10 / §11.4.65 / §11.4.44.

SHORT variant (one paste, first sentence)

Read RESUME.md then docs/CONTINUATION.md, run `git fetch --all --prune --tags`, and continue the test-suite remediation on main at HEAD 66d6fb29. **Everything is UNCOMMITTED in the working tree** (~470 files: 431 gofmt-only + 30 prior remediation + this cycle's fixes, incl. the new tests/precondition/port_conflict_ownership_test.go). The full sweep is **NOT green** — the tag is blocked. **The whole Helix platform is STOPPED and systemd-DISABLED as of 2026-07-27 evening — nothing auto-starts; you must boot infra before any infra-dependent test** (see “Session 2026-07-27 evening” below). Do NOT trust any “package ok” without checking the ambient-contention caveat in “Traps”. Next action: settle the competing-orchestrator decision (task #9) — until it is settled, postgres:15432 cannot stay up and the precondition gate cannot pass. Force-push is absolutely forbidden (§11.4.113).

FULL variant (paste-ready block)

You are resuming the HelixCode test-suite remediation cycle. The prior session landed real fixes with captured RED → GREEN evidence, and also made two attribution errors that are corrected below — read the corrections before trusting any earlier summary.

Read first: RESUME.md (this file), docs/CONTINUATION.md. Then `git fetch --all --prune --tags` and `git log --oneline HEAD..@{u}`.

State: meta-repo main @ 66d6fb29; helix_agent @ 36597718; helix_qa @ a729b51; constitution checkout @ b00ab1c. NOTHING committed, nothing tagged, nothing pushed.

Terminal goal: green full sweep → tag helix-code-1.2.0-dev-0.0.1 on the main repo AND every owned submodule carrying changes, identical helix-code- prefix (§11.4.151), fast-forward-only to all upstreams. The operator authorised the tag **conditional on the sweep being green**. It is not green. Do not tag.

Anti-bluff is binding (§11.4): every closure needs pasted runtime output; every gate touched needs a paired §1.1 mutation; no assertion may be weakened and no bare t.Skip() added to force green.

Live-state anchors

Key	Value
Meta-repo HEAD	66d6fb29 (branch main, 0 behind upstream — verified this session, HEAD . .@ empty)
helix_agent	36597718
helix_qa	a729b51
constitution (checked out)	b00ab1c — meta-repo pins 7295a189, 79 commits behind (clean FF)
Release prefix	helix-code (from HELIX_RELEASE_PREFIX in .env, mode 0600)
Latest existing tag	helix-code-1.1.0-dev-0.0.3 → next is helix-code-1.2.0-dev-0.0.1
Working tree	~470 modified files, ALL uncommitted
Sweep logs (this cycle)	qa-results/full_retest/ {helix_code_inner, helix_agent, helix_qa, verify_rules, verify_ca_20260727T13*.log
Pre-gofmt tree backup	<scratchpad>/helix_agent_pre_gofmt.patch (3322 lines)

Operator decisions recorded 2026-07-27 (binding)

1. **Ports** — do NOT hand-assign. Use the **containers submodule's dynamic-port + service-discovery** capability; extend that submodule for anything missing (§11.4.74 extend-don't-reimplement) and cover the extension with all supported test types. All systems/services/infrastructure choose an available port at runtime, bind it, and expose it through discovery; services discover each other. This supersedes the earlier “which service moves off 8100” framing.
 2. **Constitution pin** — advance to the checked-out HEAD **in this release**.
 3. **Moving baselines** — fix the tests so they stop writing to tracked files (root-cause fix, not gitignore).
 4. **Compose coupling** — **relocate mcp_servers** under `helix_agent` so no upward path traversal is needed.
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Landed this session, with captured evidence

- **gofmt across our Go code:** 507 → 0 unformatted. Vendored/archival trees deliberately EXCLUDED and reverted (see Correction 2). 431 format-only files.
 - **Production nil-deref crash FIXED** — `internal/llm/llamacpp_provider.go` `GenerateStream.req, _ := http.NewRequestWithContext(...)` discarded the error, so the next line dereferenced a nil `*http.Request`. Fault address `0x38 == offsetof(http.Request.Header)`, confirming it exactly. Guarded by `internal/llm/llamacpp_stream_malformed_url_test.go` (§11.4.115 RED_MODE polarity test). **Proven:** RED_MODE=1 PASS pre-fix / FAIL post-fix; RED_MODE=0 FAIL pre-fix / PASS post-fix; whole package ok 69.789s.
 - **G8 gate reconciled** (§11.4.120) — was a FALSE POSITIVE, see Correction 3. Now PASS, with a 7-assertion paired mutation.
 - **G13 now 82/82** — four honest Sources-verified footers, zero invented URLs, two recording explicit non-verification gaps.
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Session 2026-07-27 evening — additions, corrections, and shutdown state

PLATFORM IS DOWN AND DISABLED. All six user units stopped and **systemctl --user disabled:** `helix.target`, `helixagent.service`, `helixcode-server.service`, `helixcode-infra.service`, `helixllm-gateway.service`, `helixllm-coder.service` (+ the generated `helix-{pg,redis}-data-volume.service`). Verified `active=inactive` `enabled=disabled` on all six;

ports 7061 / 8081 / 8100 / 15432 / 16379 / 18081 / 5433 / 6380 / 11434 / 18434 / 8001
all confirmed unbound. **Nothing auto-starts on next login** — re-enable
deliberately before expecting any service.

Untouched by design (other projects on the shared host, §11.4.174): `helixterm-*`
+ `77c45f5c104d-infra`, `penpot-*`, `helix_sonarqube*`, and **atmosphere-aosp-build** (was running at shutdown — do not assume the host is idle).

Landed — PORT_CONFLICT ownership fix (RED → GREEN captured)

tests/precondition/containers_boot_test.go + new tests/
precondition/port_conflict_ownership_test.go. The gate held **two**
opposite §11.4.201 defects:

- **Fail-closed**: probed `{port, port+10000}` and declared `PORT_CONFLICT` if both answered, with zero ownership check. Our stack publishes postgres on `15432 = 5432+10000`, so *any* foreign postgres on the standard port (here `helixterm`) produced a false failure.
- **Fail-open**: `findContainerInstances` ran `docker compose ps --filter name=...` (because `detectContainerRuntime` returns `args ["compose"]`) — not a container-listing command. It returned nothing, so the duplicate-instance check **silently never fired**.

Replaced with runtime-driven ownership: `listRunningContainers` → `ourServiceInstances` (filters the `helixagent-` prefix) → `portConflict`.
Evidence: `RED_MODE=1` reproduces (*“legacy blind probe flags a foreign listener as our own duplicate”*), `RED_MODE=0` GREEN, 3/3 both polarities, ok
`dev.helix.agent/tests/precondition 0.005s`. The real-duplicate case is still caught (§11.4.120 — not weakened to a tautology).

Correction 4 — gate status in the 1334Z log is STALE

Fresh sweep `qa-results/full_retest/`
`verify_rules_20260727T184002Z.log`: **G8 PASS, G12 PASS, G13 PASS (82/82 footerd)**. Only **G7** fails, and at **18** violations, not 24. Re-read that log before acting on any earlier gate claim.

Correction 5 — the discovery “failures” were never code defects

`internal/discovery TestAllocatePort_RangeExhausted_WithEphemeral /`
`TestConcurrentAllocations / TestConcurrentReleases` pass on a quiet host:
ok `dev.helix.code/internal/discovery 0.013s`. The 1332Z run launched
the `helix_code` (133220Z) and `helix_agent` (133218Z) sweeps **2 seconds apart**;
`helix_agent`’s stress/chaos saturated the ephemeral range 32768-60999, so the

tests' preferred ports 50000-50049 were all unavailable (forcing fallback into the 19-port api range → exactly the 16 unique ports logged) and `net.Listen(":0")` returned `EADDRINUSE`. Independent confirmation of Correction 1.

NEW blocker — competing container orchestrators (task #9, architectural)

`helixagent.service`'s container adapter recreates `helixagent-postgres` from **`docker-compose.yml`, where postgres has NO ports: section** — so 15432 is never published. Manual `docker-compose.test.yml` runs `DO publish` it. With both in play postgres crash-loops: could not bind IPv4 address "0.0.0.0": Address in use (host 5432 is held by the unrelated `helixterm` stack). **While the service runs, the precondition gate cannot see 15432.** Compounding it, `scripts/ensure-infrastructure.sh` health-checks `${DB_PORT:-15432} / ${REDIS_PORT:-16379}` but boots the default compose — it cannot satisfy its own health check (task #7).

Recommended (evidence-determined, reversible, NOT yet applied — needs the operator's call because it changes which stack owns the live platform's containers): point the infra boot at `docker-compose.test.yml` for postgres/redis, restoring the script's own declared port contract.

Sweep results 2026-07-27 — NOT GREEN

Stream	Verdict	Detail
helix_qa	GREEN	142 ok / 0 FAIL / 23 no-test-files
governance cascade	GREEN	277 PASS / 0 FAIL; 227 anchors discovered, all 6 carriers to §11.4.234
helix_code inner	NOT-GREEN	192 ok / 3 FAIL / 0 build-failed (285 pkgs, 0 cached)
helix_agent	NOT-GREEN	355 ok / 23 FAIL (506 pkgs); tests/ integration killed by 10-min timeout → PARTIAL
constitution rules	NOT-GREEN	16 gates, 3 FAIL (G7/G8/G13), G14 SKIP

Stream	Verdict	Detail
		(docs_chain engine absent)

G8 + G13 are now fixed. **G7 remains failing by design** (see below).

CORRECTIONS — read these before trusting earlier summaries

Correction 1 — three concurrent sweeps contaminated the results (§11.4.119).

The prior session ran three full `go test ./...` sweeps simultaneously on one host. Captured proof of contamination: `112× cannot assign requested address` and `8× bind: address already in use`, with sibling sweep PIDs verified by `cwd/argv`. Test sweeps that bind ports are NOT parallelisable — partition by RESOURCE, not by repo. **Several reported failures are ambient-suspect, not proven code defects.** A serial, host-quiet re-run is REQUIRED before any of them are called real.

Correction 2 — docs/research/go-elder-plinius-v3 is NOT vendored. It has no nested `.git` and no LICENSE; the nested repos under `docs/research/` belong to SIBLING directories (Gandalf-Solutions, CL4R1T4S, LEAKHUB, AutoRedTeam, CLAUDE-CODE-SYSTEM-PROMPT). Reverting `gofmt` there was still CORRECT, but for a different reason: `go list ./...` returns **0** packages from it — it is archival research intake outside the root module graph that no `build/vet/test` gate can validate.

Correction 3 — G8 was a false-positive gate, not a documentation defect. Both items already carried complete, truthful Obsolete-Details. The gate asserted `Triple-check::canonical (constitution/Constitution.md:7687)` and the generator (`obsolete.go:187`) both say `Triple-check evidence::`. That markdown is machine-generated from the SQLite SSoT — hand-editing it would have broken **G11** (byte-identical `md → db → md` round-trip) and been reverted at next regeneration.

G7 (§11.4.83) — 24 violations split THREE ways

- **Class C (6) — FALSE violations.** Evidence exists; the gate cannot match it. It requires the COMMIT SUBJECT to contain the evidence directory basename, but run-ids are timestamped slugs (`hxc135_20260712T130921Z`). Affected: `3e67fa1a`, `54a76c3c`, `ca76e14b`, `7210f373`. Two more (`02e9e505`,

1254e0a6) committed evidence as FLAT docs/qa/*.md files, invisible because the gate enumerates only directories.

- **Class B (3)** — genuine non-features tripping the heuristic: d6c05f76 (README only), f9dcf6a6 (doc-comment only), b058c7c2 (const → var test-robustness, prod default unchanged).
- **Class A (15)** — genuine features shipped with NO evidence anywhere, **including 3 security fixes**: 4727a9d0 (CORS wildcard+credentials), 9c876819 (CSWSH /ws auth), 2ff55c31 (wire-facade 401). Full list: fbfffd7d 0e3bb747 6efadd15 225cdf77 aa6b20b4 eb233785 f8c38181 67c9a9bc 4727a9d0 9c876819 2ff55c31 a21ad7ca 51c058b1 66f9c21e c9bad26a.

HARD CONSTRAINT: the [no-qa-evidence] opt-out lives in the COMMIT MESSAGE and all 24 are already pushed. Retroactive application requires history rewrite — **forbidden absolutely** (§11.4.113 / CONST-043). The opt-out is available only going forward.

Class A is an OPERATOR DECISION, not yet made: retrospective verification runs against real infra (legitimate — the transcript records a real run performed now; security fixes first) VS a documented baseline bump like the gate header records for 118 commits on 2026-06-22. Fix Class C FIRST — bumping before that hides a real matching bug and overstates the debt by 6.

Open defects — confirmed, NOT yet fixed

- **IPv6 bracket-unawareness — TWO independent bugs, one class, neither a regression** (no shared helper; both non-bracket-aware since creation):
 - helix_code/internal/llm/response_err_round53_test.go:869 mustSplitHostPort re-joins the output of net.SplitHostPort (which strips IPv6 brackets by design) with plain concatenation → http://::1:PORT.
 - helix_agent/internal/vectordb/qdrant/client_mock_test.go:31-35 uses strings.Split(url, ":") on a bracketed authority → host == "[", and a discarded Sscanf error silently defaults the port to 80 → http://[:80.
 - Trigger for both: httptest.NewServer's IPv6 fallback when IPv4 loopback bind fails. **These tests are environment-flaky by construction.**
- **i18n key leak — LATENT, not live.** tool_schema ships a consumer-injected translator seam; **nothing anywhere calls toolschema.SetTranslator**, so activeTr stays NoopTranslator and tr() echoes the key. Bundle entry EXISTS and is correct (active.en.yaml:50 → "Create git commit"). No production path currently reaches it, so no user sees it — but it arms on first use, and all 36 tr() sites include user-visible ToolResult.Error payloads.

Do NOT “fix” the test to expect the raw key — that cements the unwired state and disarms the only cross-module check.

- **HXC-131 evidence path is dead** — docs/qa/followup_fixes_20260712T085616Z/HXC131_evidence.md never existed (that commit added only HXC133_evidence.md). Genuine evidence IS at tracked scratch/discovery/fixes/HXC131_evidence.md. Correct repair is a DB mutation on the SSoT (workable-items obsolete-details -- evidence ... + re-render), not a doc edit.
 - **sdk/web/node_modules/flatted/golang/pkg/flatted/flatted.go** — genuinely third-party vendored code sitting INSIDE the gofmt gate. Clean today; an npm bump reintroduces the tension.
 - **go vet ./... covers only the root module** — 36 nested modules (ours included) are never vetted. Real coverage hole, own item.
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In flight at handoff (§11.4.147 — a crashed agent is NOT complete)

Four background agents dispatched; results NOT yet collected or verified: 1. Independent review (§11.4.142) of the llamacpp fix + its RED test. 2. G7 Class-C matching-bug reconciliation + paired mutation. 3. Extend the discarded-error guard to 4 sibling sites (internal/verifier/embedded_server.go:84, together/client.go:72, replicate/client.go:87,129), TDD RED-first. 4. Apply the prepared carrier elision patch (48 entries, 6 carriers) + verify §11.4.157 lockstep + regenerate exports.

If any did not land, re-dispatch — do not assume completion.

Traps that already cost time — do not re-learn these

- **A “package ok” may be ambient, not repair.** internal/llm reported ok after the crash fix — including 3 round-53 tests that had failed. Those pass because httptest bound IPv4 that run, NOT because the IPv6 defect was fixed. It is unfixed. Green here is environment.
- **The legacy summary generators DESTROY data.** scripts/generate_{issues,fixed}_summary.sh read the TEXT trackers; committed summaries are SQLite-derived (§11.4.93/.95). Running them rewrites 344 items down to 188 — 156 items of tracked state destroyed.
- **make test can never pass on a headless host** — runs go test ./... with no -tags=nogui.

- A “completed / exit 0” notification describes the WRAPPER, not the work. Verify the artefact — mtimes, pgrep, pasted output.
- **replace_all is unsafe when a NEW helper’s body contains the pattern being replaced** — rewrites the helper into infinite self-recursion.
- **PDF export silently drops wide tables’ rightmost column** (§11.4.168) while reporting rendered=2 failed=0. Keep load-bearing evidence out of wide tables.
- **git ls-files '*.go' | xargs gofmt -w reaches vendored AND archival trees.** Always exclude docs/research/ (and check node_modules) before formatting.
- **podman pod rm -f pod_helix_agent KILLS helixagent.service.** Its container adapter sees the pod vanish and shuts down gracefully ("Shutting down container adapter...", status=0/SUCCESS) — it looks like a clean exit, not a casualty. Cost this session: chaos/api + chaos/auth regressed ok → connection refused on :7061, and a whole 30-min sweep ran against a dead server. Stop the service first, or don’t touch its pod.
- **pgrep -f 'go test' matches YOUR OWN shell** — the loop’s command line contains the literal string. It killed my own bash mid-command (exit 144). Always exclude \$\$/\$PPID and verify by cwd (§11.4.174 / §11.4.201 carrier footgun; §12.12 names this exact class).
- **:0 failing with EADDRINUSE is the ephemeral-exhaustion tell.** Binding port 0 asks the kernel for *any* free port; it can only fail when the range is genuinely exhausted. That single line is the fastest way to distinguish host contention from a real allocator bug.
- **The test stack and the platform stack use DIFFERENT ports.** Precondition demands 15432/16379/18081 (docker-compose.test.yml); the live platform uses 5433/6380 (helixcode-infra-*). Seeing 5432/6379 listening proves nothing about test readiness — those belong to another project.

Sweep commands (exact) — RUN SERIALY, ONE AT A TIME

```
# Host must be quiet. Three of these in parallel exhausts IPv4
# ephemeral
# ports, forces httptest onto its IPv6 fallback, and produces
# failures that
# are contention artefacts rather than code defects (§11.4.119).
cd helix_code                && go test -tags=nogui -count=1 ./...
cd submodules/helix_agent    && go test -count=1 ./...
cd submodules/helix_qa       && go test -count=1 ./...
./scripts/verify-all-constitution-rules.sh
./scripts/verify-governance-cascade.sh
```

-tags=nogui is REQUIRED for the inner module (headless host, no X11/GL dev headers). -count=1 is REQUIRED — an earlier run reported 250 ok of which 249 were (cached).

Binding constraints (restated — §11.4.131(C))

- **Anti-bluff §11.4** — every PASS needs captured runtime evidence; metadata-only / config-only / absence-of-error / grep-without-runtime PASS are critical defects.
- **§11.4.113 — force-push is absolutely forbidden**, no exception, no operator override. Integrate by merging onto latest main, push fast-forward only.
- **§11.4.151** — tags carry the helix-code- prefix, identical across main repo and every owned submodule in one release.
- **§11.4.142** — every change needs INDEPENDENT review before commit/build; author self-verification (§11.4.92) precedes but never satisfies it.
- **§11.4.84** — account for every modified file before `git add`; unaccounted entries → ABORT. Never `git add -A` here.
- **§11.4.119** — one owner per exclusive resource; partition parallel work by RESOURCE, not by repository.
- **§11.4.185** — manual QA-team confirmation is the final sufficiency gate; not self-certifiable.